



APDA 301 PROJECT BRIEF AND DOCUMENTATION

Secure Student Management System on Azure

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1. Project Overview

Institution: FutureTech Academy

Objective: Develop a secure, cloud-based student management system for managing student data efficiently.

Technology Stack: .NET Core, Azure Cosmos DB, Azure Blob Storage, Azure App Services

2. System Architecture

Web Layer: Built using .NET Core to handle UI and logic.

Database Layer: Azure Cosmos DB stores student records.

Storage Layer: Azure Blob Storage stores images in 'student-images'.

Integration: profileImageUrl links database and images.

3. authentication & Security

OAuth 2.0 for secure login (Google/GitHub).

Role-Based Access Control (RBAC) for user permissions.

SAS tokens for secure image access.

4. Features

CRUD operations: Add, view, edit, delete students.

File validation: Only JPEG/PNG allowed.

User-friendly dashboard interface.

Deployment using Azure App Services with deployment slots.

5. Deployment & Maintenance

Live link: [FUTURETECH ACADEMY LIVE PROJECT LINK](#)

[GitHub source code Link](#)

Monitoring via Azure Application Insights.

Auto-scaling enabled.

6. Testing

Unit testing for CRUD and authentication.

Security testing for OAuth and RBAC.

Performance testing for response times

7. Screenshots Section

7.1 First-Time Setup (New Deployments Only)

If the system is newly installed, the first user must configure the Primary Administrator account.

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First Time Setup

Create the admin account for FutureTech Academy

This page appears once. Your credentials are stored securely via BCrypt hashing in Azure Cosmos DB.

YOUR NAME
John Smith

ADMIN EMAIL
admin@example.com

PASSWORD

CONFIRM PASSWORD

Create Admin Account

Enter your **Full Name**, **Admin Email**, and a secure **Password**.

Click **Create Admin Account**.

7.2 Regular Sign-In

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FutureTech Academy
Student Management Portal

ADMIN SIGN IN

Email
admin@example.com

Password
••••••••

Sign In

or

 Continue with Google

Signing in with Google gives you **Viewer** access. Contact the admin to request elevated permissions.

Admin Login: Enter your registered email and password on the main Sign-In screen.

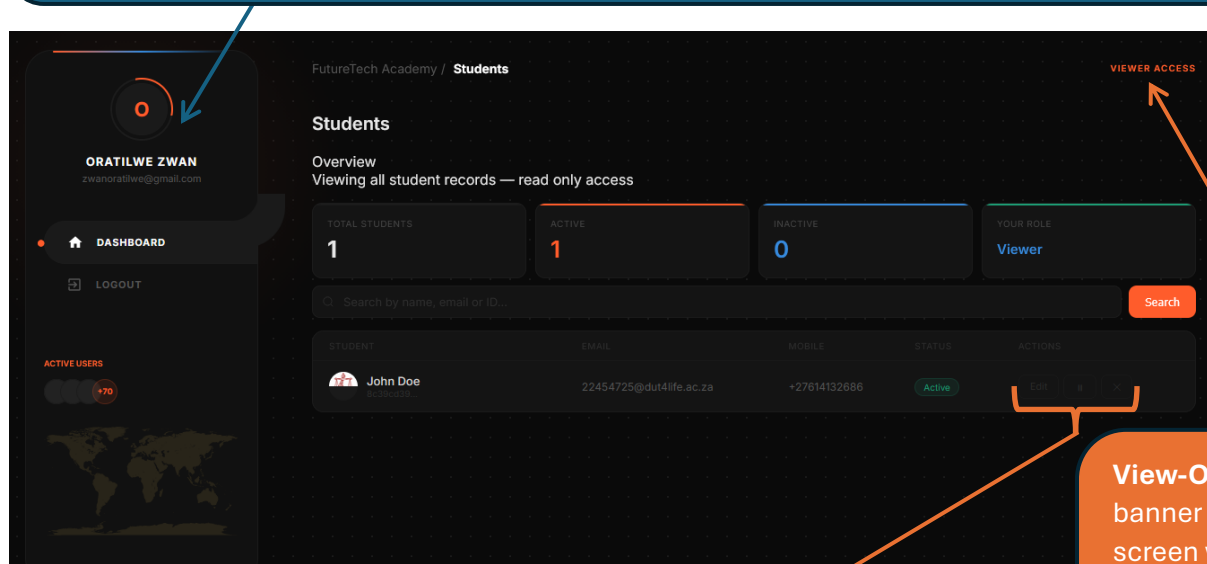
Google Sign-In: Click **Continue with Google**.

- If you are an Admin, you will be granted full access.
- If you are a new user, you will be granted **Viewer Access** by default.

7.3 Viewer Access (Read-Only)

Once logged in, you will see the sidebar and main workspace. The sidebar contains:

- **User Profile:** Shows your name, email,
- **Navigation:** Switch between the **Dashboard** (Student List) and **Students** (Add New). (Admin)



View-Only Banner: A banner at the top of the screen will confirm "Viewing all student records — read only access."

Buttons to edit, change student status and delete student

Editing Student Records

1. Locate the student in the Dashboard list.
2. Click the **Edit** button in the **Actions** column.
3. Update the necessary fields and save changes.

Deactivating/Activating a student

To temporarily suspend a student without deleting their data:

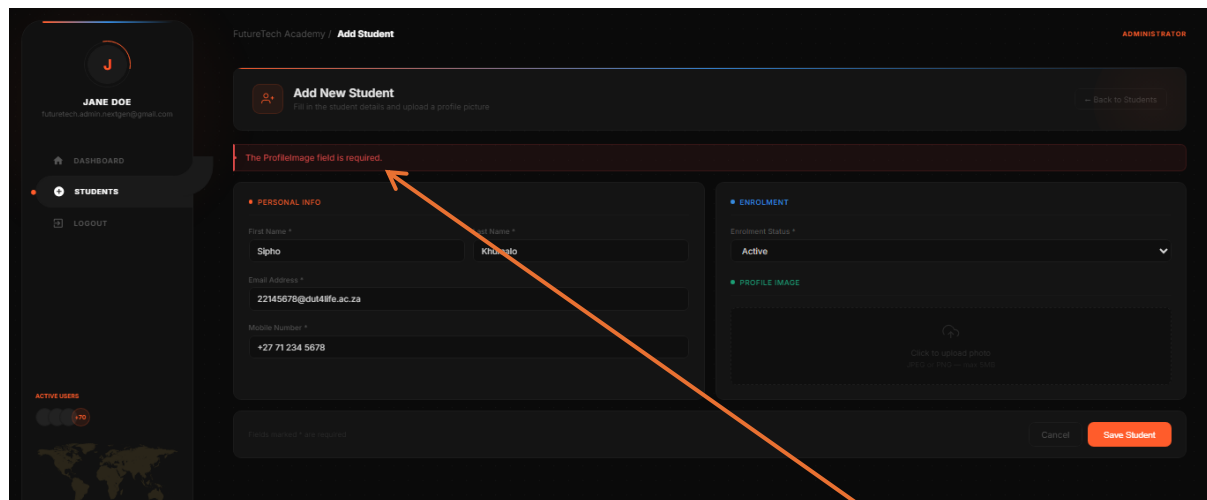
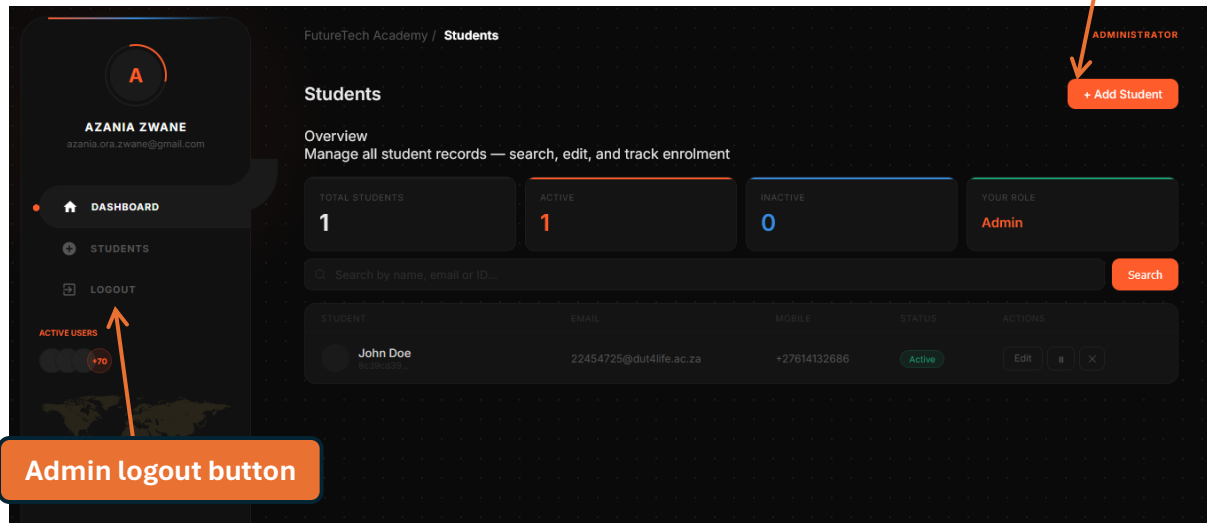
1. Click the **Pause (II)** icon in the Actions column to set them too **Inactive**.
2. Click the **Play** icon to return them to **Active** status.

Deleting a student

1. Click the **Delete (X)** icon in the Actions column.
2. Confirm the deletion when prompted. *Warning: This action is permanent no rollback.*

7.4 Administrator Functions

Add Student button located at the top right of the Dashboard.

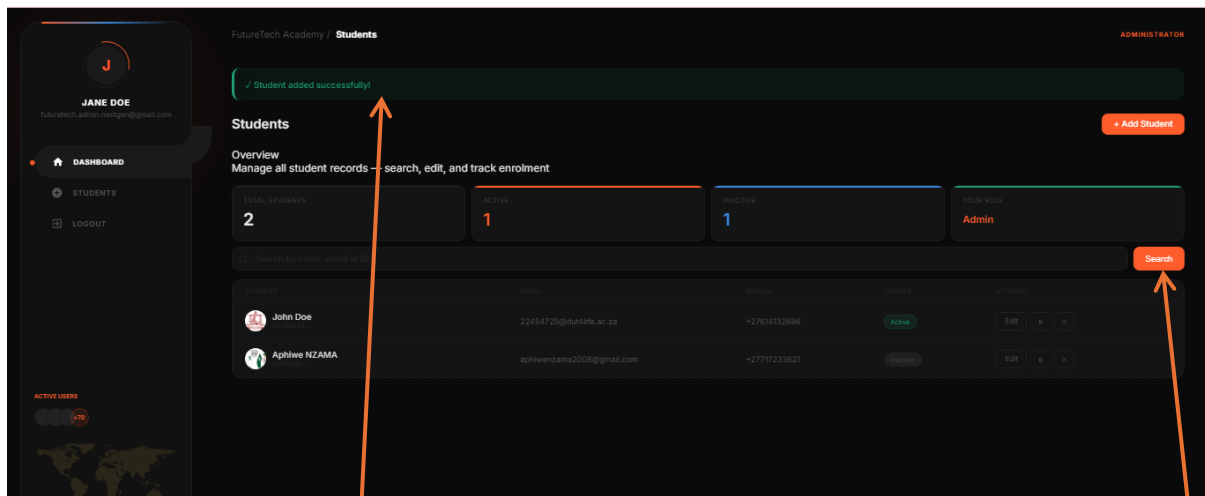


After Clicking

1. **+ Add Student** button located at the top right of the Dashboard.
2. This form will appear to add student information Personal Info
3. Fill in the First Name, Last Name, Email, and Mobile Number.
4. **Enrolment:** Select the status (Active/Inactive) from the dropdown.
5. **Profile Image:** Click the upload area to select a student photo (JPEG/PNG).

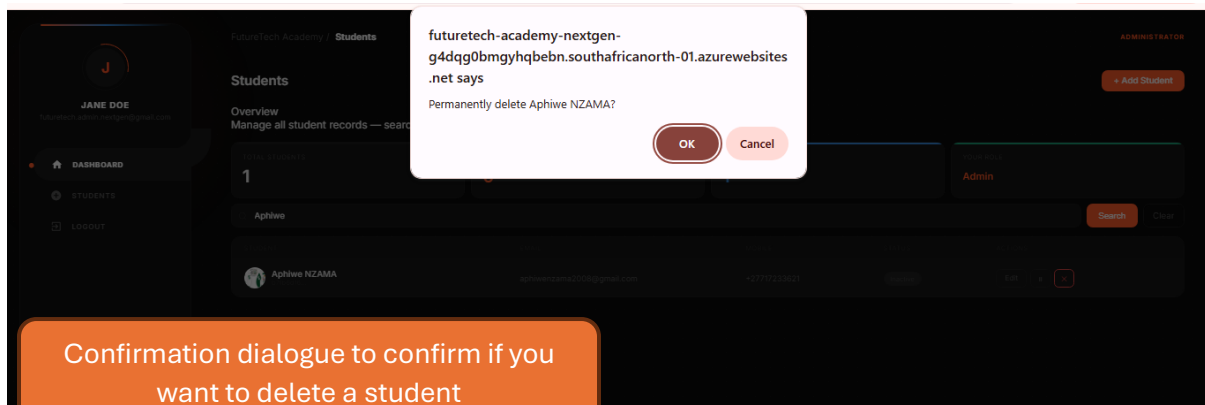
error message if the information entered is incorrect or incomplete

7.4 Administrator Functions. continued



Search button under the overview of the Dashboard.
Search using ID, Name, Email

Confirmation message if adding a new user was successful



Confirmation dialogue to confirm if you want to delete a student

7.5 Unit testing

Test Explorer

Ready

Test	Duration	Traits	Error Message
FutureTech.Test (15)	1,3 sec		
FutureTech.Test (15)	1,3 sec		
BlobStorageServiceTests (6)	683 ms		
GenerateSasUrl_NullOrEmptyUr...	< 1 ms		
UploadImage_FileTooLarge_Fail...	1 ms		
UploadImage_FileWithinSizeLi...	1 ms		
UploadImage_InvalidFileType_T...	< 1 ms		
UploadImage_ValidJpeg_Passes...	679 ms		
UploadImage_ValidPng_PassesT...	2 ms		
StudentModelTests (9)	608 ms		

Test Explorer

Ready

Test	Duration	Traits	Error Message
StudentModelTests (9)	608 ms		
Student_DefaultId_IsNotEmpty	< 1 ms		
Student_DefaultIsDeleted_IsFalse	427 ms		
Student_InvalidEmail_FailsValida...	< 1 ms		
Student_MissingEmail_FailsValid...	< 1 ms		
Student_MissingFirstName_Fails...	< 1 ms		
Student_MissingLastName_Fails...	179 ms		
Student_MissingFirstName_FailsValidation			
Student_SoftDelete_SetsCorrect...	1 ms		
Student_ValidData_PassesValida...	2 ms		

Test Explorer

7.6 Cosmos DB (Azure Portal)

The screenshot shows the Azure Portal interface for the 'futuretech-cosmodb' Azure Cosmos DB account. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Quick start, Data Explorer, Mirroring in Fabric, Container Copy, Resource visualizer, Settings, Integrations, and Containers. The main content area displays the account overview, including a notification about the April 28th Cosmos DB Conf 2026. Below this, the 'Essentials' section provides key account details in a table:

Property	Value
Status	Online
Read Locations	South Africa North
Write Locations	South Africa North
URI	https://futuretech-cosmodb.documents.azure.com
Free Tier Discount	Opted Out
Capacity mode	Serverless

Additional details include Resource group (future-tech-rg), Subscription (Azure for Students), Subscription ID (b4ce6869-21b0-44d9-bd44-676a95a99ddd), and Total throughput limit (No total throughput limit). A 'See more' link is provided for further details. Below the Essentials section, there are tabs for Setup & Optimization, Learning Resources, Containers & Data, Monitoring, and Recommendations. The Setup & Optimization section includes links for Load & Query Data, Essential Design Decisions, Build & Test, and Get Production Ready.

7.7 Blob Storage Container

The screenshot shows the Azure Portal interface for the 'futuretechstoragedb' Storage account. The left sidebar contains navigation links: Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Storage Mover, Partner solutions, Resource visualizer, and Data storage. The main content area displays the account overview, including a notification about preventing unauthorized access. Below this, the 'Essentials' section provides key account details in a table:

Property	Value
Resource group (...)	future-tech-rg
Location	southafricanorth
Subscription (move)	Azure for Students
Subscription ID	b4ce6869-21b0-44d9-bd44-676a95a99ddd
Disk state	Available
Performance	Standard
Replication	Locally-redundant storage (LRS)
Account kind	StorageV2 (general purpose v2)
Provisioning state	Succeeded
Created	2026/04/11, 23:33:47

Below the Essentials section, there are tabs for Properties, Monitoring, Capabilities (7), Recommendations (1), Tutorials, and Tools + SDKs. The Properties section includes a 'Blob service' section with settings for Hierarchical namespace (Disabled) and Default access tier (Hot). The Security section includes settings for Require secure transfer for REST API operations (Enabled) and Storage account key access (Enabled).